What is Seaborn?

Seaborn is a Python data Visualization library based on Matplolib. It provides a high-level interface for drawing attractive and informative statistical graphics.

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
tips = sns.load_dataset('tips')
```

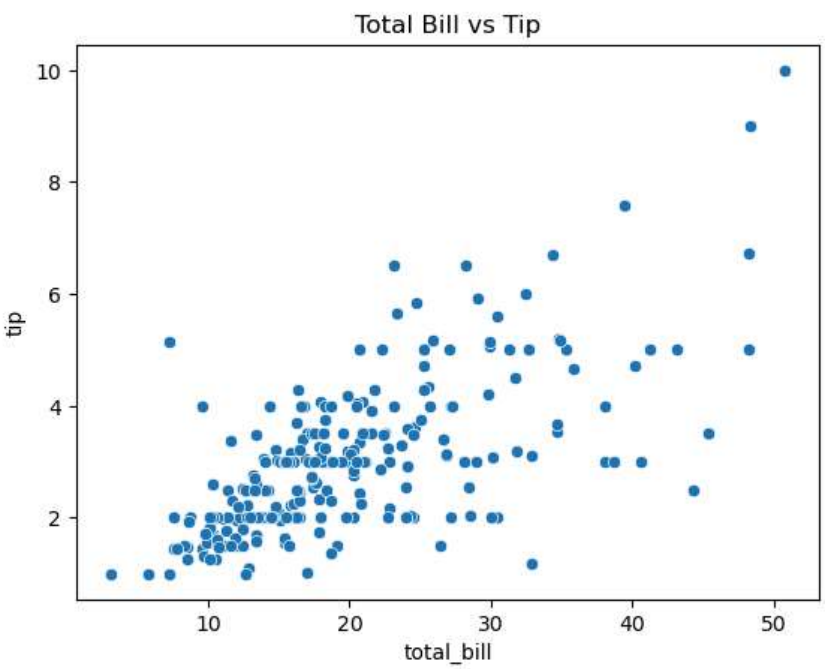
```
tips.head()
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

Scatter plot

A scatter plot displays values for two variables as a collection of points.

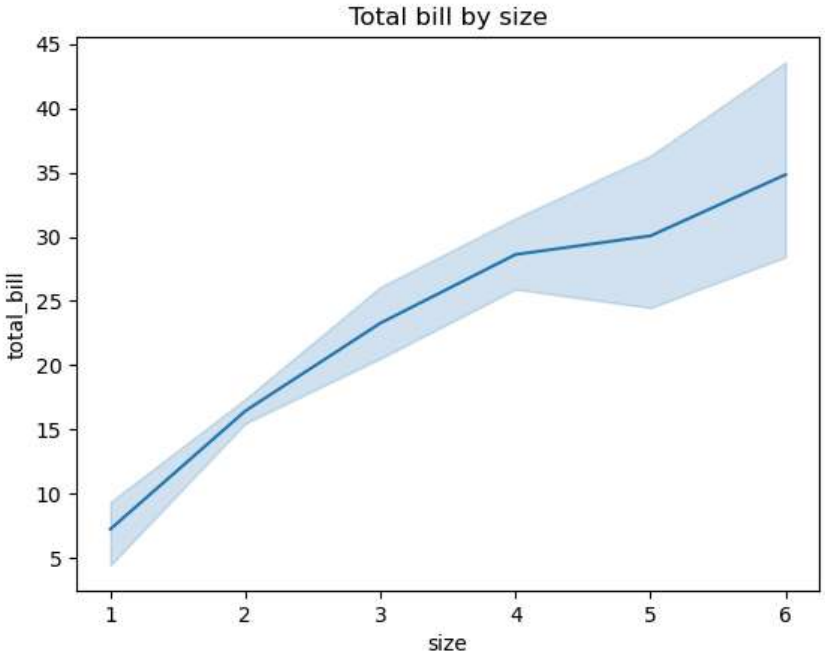
```
sns.scatterplot(data=tips, x='total_bill', y= 'tip')
plt.title('Total Bill vs Tip')
plt.show()
```



Line Plot

A line plot display infomation as a series of data points connected by a straight line segments.

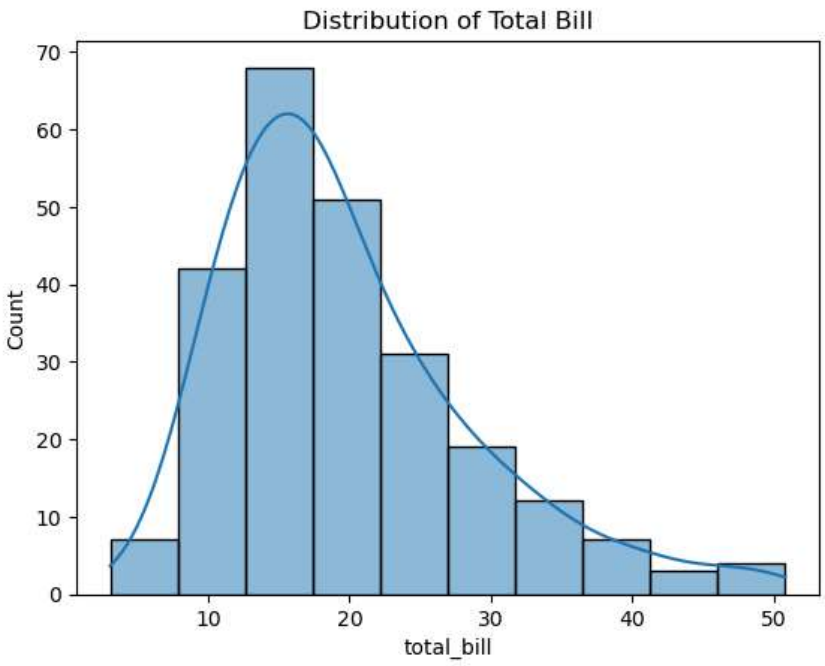
```
#line plot
sns.lineplot(data=tips, x='size', y='total_bill')
plt.title('Total bill by size')
plt.show()
```



Histogram

A histogram is an accurate representation of the distribution of numerical data.

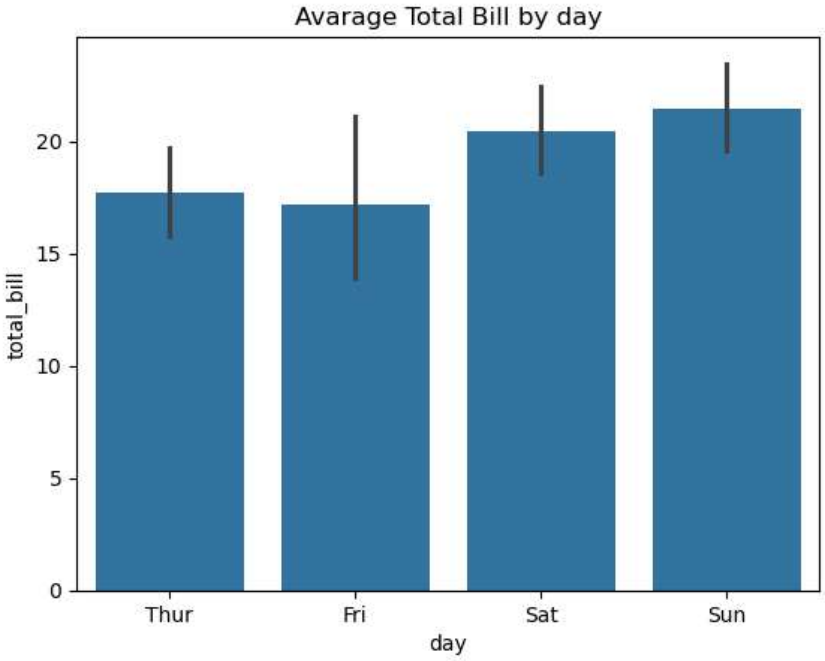
```
#histogram
sns.histplot(data=tips, x='total_bill', bins=10, kde= True)
plt.title('Distribution of Total Bill')
plt.show()
```



Bar Plot

A bar plot is a way of summarizing a set of categorical data.

```
# Barplot
sns.barplot(data=tips, x='day', y='total_bill')
plt.title('Avarage Total Bill by day')
plt.show()
```



Start coding or generate with AI.

```
# Import seaborn
import seaborn as sns

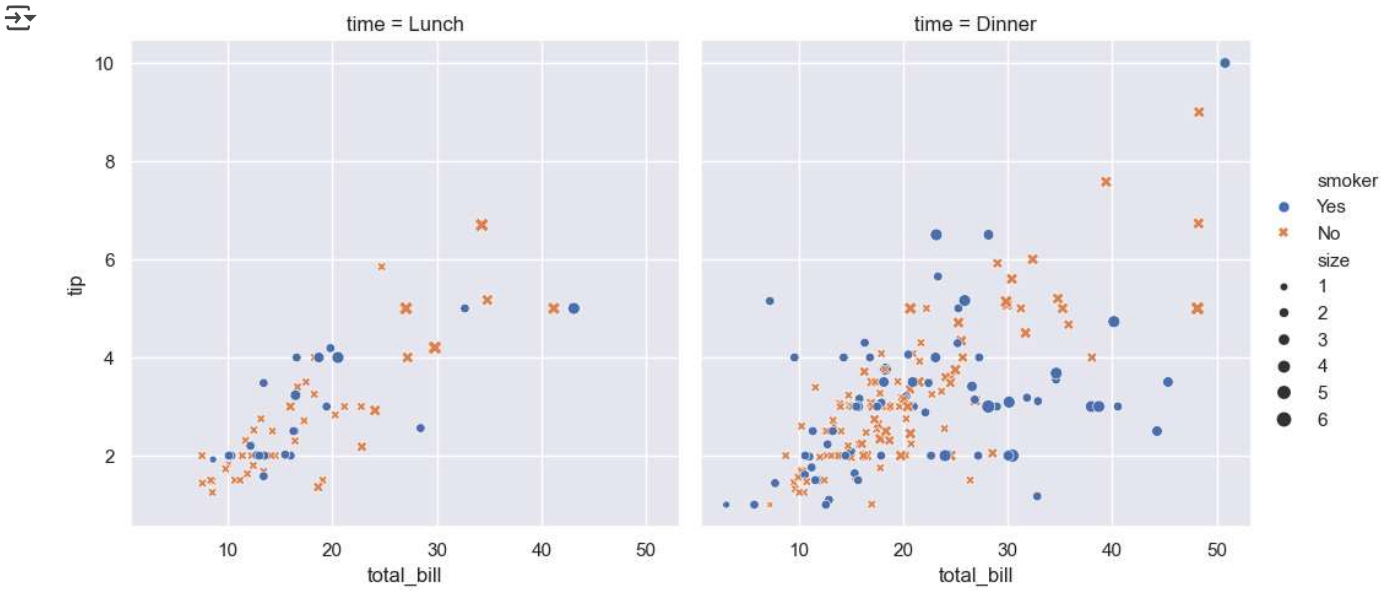
# Apply the default theme
sns.set_theme()

# Load an example dataset
tips
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3
240	27.18	2.00	Female	Yes	Sat	Dinner	2
241	22.67	2.00	Male	Yes	Sat	Dinner	2
242	17.82	1.75	Male	No	Sat	Dinner	2
243	18.78	3.00	Female	No	Thur	Dinner	2

244 rows × 7 columns

```
# create a visualization
sns.relplot(data = tips,
            x='total_bill', y='tip',col='time',
            hue ='smoker', style='smoker', size='size'
)
plt.show()
```




```
dots = sns.load_dataset('dots')
sns.relplot(data=dots, kind='line',
            x='time',y='firing_rate', col='align',
             hue='choice', size='coherence', style='choice',
             facet_kws =dict(sharex=False)
            )
plt.show()
```




```
dots
```

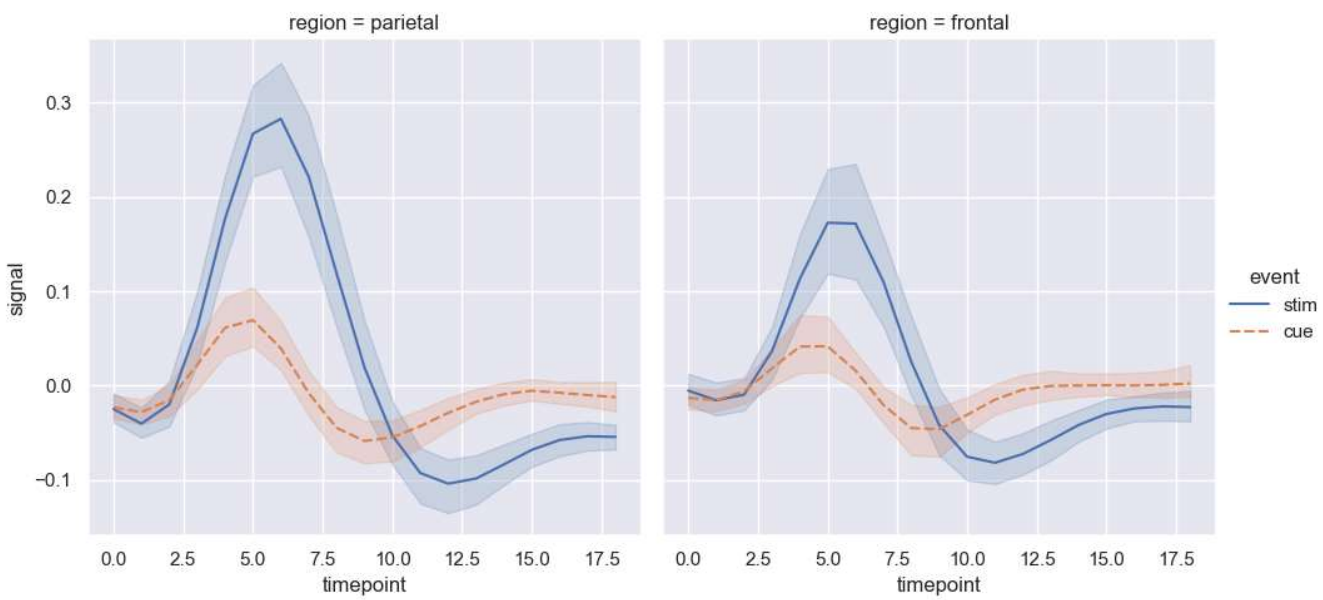
	align	choice	time	coherence	firing_rate
0	dots	T1	-80	0.0	33.189967
1	dots	T1	-80	3.2	31.691726
2	dots	T1	-80	6.4	34.279840
3	dots	T1	-80	12.8	32.631874
4	dots	T1	-80	25.6	35.060487
...	...	...	...	...	...
843	sacc	T2	300	3.2	33.281734
844	sacc	T2	300	6.4	27.583979
845	sacc	T2	300	12.8	28.511530
846	sacc	T2	300	25.6	27.009804
847	sacc	T2	300	51.2	30.959302

848 rows × 5 columns

```
fmri = sns.load_dataset('fmri')
import pandas as pd
import numpy as np

fmri.replace([np.inf, -np.inf], np.nan, inplace=True)


sns.relplot(data=fmri, kind='line',
           x='timepoint', y='signal', col='region',
            hue='event', style='event'
           )
plt.show()
```



Statistical estimation in seaborn goes beyond descriptive statistics. For example, it is possible to enhance a scatterplot by including a linear regression model (and its uncertainty) using lmplot()

```
sns.lmplot(data=tips, x='total_bill', y='tip', col='time', hue='smoker')
```

```
<seaborn.axisgrid.FacetGrid at 0x24bdf7b6350>
```

